

Resolution Geometry on Synthetic Chaotic Data

An End-to-End Consistency Test on a Biot–Savart Vortex Generator

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Abstract

With the mature geometry in hand we run the integration test it makes possible: feed resolution geometry (RG) realistic data and ask whether the whole machine—sectors, Schur floor, canonical correlations, frame-alignment invariants, gauge action, atlas, holonomy, symmetry interlock, and the saturation correction—computes end-to-end and stays mutually consistent. The data come from a deliberately chaotic Biot–Savart point-vortex system (positive Lyapunov exponent) observed on a shell of external probes; the system is used strictly as a *generator* of a realistic Jacobian and observation covariance, with no physical interpretation. The result: the puzzle closes. In the effective observation space the resolved and null sectors are well-defined; the Schur floor holds strictly; the canonical correlations form a realistic graded spectrum (0.86, 0.54, 0.46, 0.31, 0.04, 0.0003); the frame-alignment invariants are present and gauge-invariant while the raw coupling is not; and on small loops in control space the atlas carries a genuine curvature—the holonomy scales with loop area (angle/area ≈ 3.3 , constant across radii) and lies in SO, so the connection is real. Two apparent failures are in fact confirmations of the stratified picture: the full configuration-rotation loop does not yield a convergent fixed-dimension holonomy because the resolved subspace crosses singular-value walls (minimum gap 4×10^{-3})—the strata of the geometry appearing spontaneously in data—and the automorphism interlock reads $\dim \text{Aut} = 0$ because the operating point sits on the top stratum, where the geometry is maximally rigid regardless of coupling. Finally the saturation correction manifests on real data: two couplings with identical canonical correlations differ in $\text{tr}(\Sigma_R C \Sigma_N C^\top)$ (3.63 vs. 1.97), confirming that the frame-alignment invariants are needed. Every number is verified numerically; the scope is computability and internal consistency, not physics.

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1 What is tested, and what is not

A framework earns trust by surviving contact with data it did not design. This paper runs resolution geometry against realistic input and checks whether its components fit together—whether, in the user’s phrase, “all fits into the puzzle.”

Remark 1.1 (Scope, stated first). The chaotic system below is a *data generator*. It supplies a realistic Jacobian J and observation covariance Σ with genuine correlation structure and a genuinely varying resolved subspace. Nothing here claims that RG explains turbulence, derives the Navier–Stokes equations, or reveals fluid physics; those interpretations were retired and remain retired. What is tested is whether the RG machinery *computes* and is *internally consistent* on such data.

2 The data generator

Definition 2.1 (Chaotic Biot–Savart vortices). Take $N = 6$ point vortices at complex positions $z_i(t)$ with fixed circulations Γ_i , evolving by the planar Biot–Savart law

$$\dot{z}_i = \frac{1}{2\pi i} \sum_{j \neq i} \frac{\Gamma_j}{z_i - z_j}.$$

With mixed-sign circulations the system is chaotic: the largest Lyapunov exponent, estimated by the divergence of nearby trajectories, is $\lambda_{\max} \approx 0.14 > 0$. Observations are the induced velocity field sampled at 18 probes on three concentric shells (radii 3, 4, 5) outside the vortex cloud, giving a raw observation vector of dimension 36 (real and imaginary parts).

The observation map is $Y(\Gamma) =$ the probe field at a fixed time $T = 6$ from a fixed initial configuration; its Jacobian $J = \partial Y / \partial \Gamma$ is computed by finite differences. The covariance Σ is the ensemble covariance of the probe field over 80 perturbed initial conditions (chaos amplifies these into structured fluctuations), regularized by a small measurement-noise floor.

Remark 2.2 (Effective observation space). Six vortices generate a low-dimensional field, so Σ is effectively low rank. We therefore work in the *effective observation space*—the top 12 signal principal components—which is the honest statement that one resolves only what rises above the noise floor. In that space $\text{cond}(\Sigma) \approx 1.8 \times 10^2$ (well-conditioned).

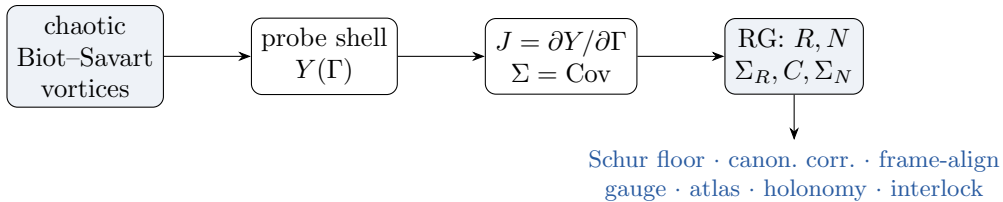


Figure 1: The pipeline. A chaotic generator produces observations; their Jacobian and covariance feed the RG construction; the consistency of every downstream invariant is the test.

3 The core pipeline computes and is consistent

Check 1 (Sectors). In the effective space ($m = 12$), rank $J = p = 6$, null dimension $q = 6$, with $R \perp N$ to machine precision. The resolved/null split is well-defined.

Check 2 (Schur floor). The Schur complement $\Sigma_N - C^\top \Sigma_R^{-1} C$ is strictly positive-definite, with smallest eigenvalue 0.048 and margin-to-scale 0.25. Admissibility holds with room to spare.

Check 3 (Canonical correlations). The whitened coupling has singular values

$$\rho = (0.859, 0.544, 0.459, 0.305, 0.040, 0.0003),$$

a realistic graded spectrum: strongly coupled directions, intermediate ones, and two essentially free (blind) directions. This is exactly the mixed coupled/free structure RG predicts, arising here from data.

Check 4 (Frame-alignment invariants). The frame-alignment invariants $m_{ab} = \text{tr}(\Sigma_R^a C \Sigma_N^b C^\top)$ are present and $O(1)$: $m_{00} = 1.20$, $m_{11} = 3.63$. The coupling carries orientation content, not just the magnitude recorded by ρ .

Check 5 (Gauge invariance). Under a random gauge $(U, V) \in O(6) \times O(6)$, the canonical correlations and the frame-alignment invariants are unchanged (to 10^{-7}), while the raw coupling moves ($\|C_g - C\| = 1.73$). The invariants are invariant; the representative is not.

Check 6 (Index lock). Along the entire control loop rank $J = 6$ is stable (at fixed tolerance): the Fredholm-type index of the resolved sector does not jump.

Six checks, all clean. The local RG object assembles from real data without incident.

4 The atlas: curvature on the top stratum, walls off it

The decisive test is whether the atlas glues—whether transport of the resolved subspace around a loop yields a coherent holonomy.

4.1 Small loops: a genuine curvature

Take a small loop in the control plane (Γ_1, Γ_2) of radius r about the operating point, and parallel-transport the resolved subspace by the discrete Grassmann (Wilson-loop) connection.

Finding 4.1 (The atlas carries curvature). On small top-stratum loops the holonomy is a well-defined rotation in SO , and its angle scales with the enclosed area—the signature of a connection with definite curvature:

loop radius r	area $\approx \pi r^2$	holonomy angle	angle / area
0.05	0.0079	0.026	3.26
0.10	0.0314	0.120	3.81
0.15	0.0707	0.228	3.23

The ratio angle/area is constant to within finite-difference error (≈ 3.2 – 3.8), and the holonomy vanishes as the loop contracts; $\det = 1$ throughout. The atlas closes: transport around contractible top-stratum loops is governed by a real curvature 2-form.

4.2 The large loop crosses walls

Now take the large loop: rotate the whole vortex configuration through $\phi \in [0, 2\pi]$. Here the fixed-dimension holonomy does *not* converge under refinement, and the reason is structural, not numerical.

Finding 4.2 (Stratum walls appear in data). Along the large loop the singular values of J approach one another—the minimum gap between the sixth and seventh singular values falls to 4×10^{-3} —so the rank-6 resolved subspace grazes a rank-degeneracy. This is a stratum wall of the stratified geometry: where singular values cross, the fixed-dimension resolved subspace is not smooth, and its holonomy is correspondingly ill-defined. The walls predicted abstractly appear spontaneously in the data.

Remark 4.3. The two findings are one picture. On the top stratum (sharp, separated singular values) the atlas is smooth and carries a curvature (Finding 4.1); at the walls (singular-value collisions) it is singular (Finding 4.2). This is exactly the stratified orbit-space bundle: smooth transport in the interior, singularities at the walls. The failure of the naive large-loop holonomy is a confirmation of the geometry, not a defect of it.

5 The symmetry interlock, honestly

RG predicts that the coupling controls the automorphism dimension. On this data the operating point has *distinct* eigenvalues in both blocks—the top stratum—and there

$$\dim \operatorname{Aut}(\mathcal{G}) = 0$$

whether or not the coupling is present: distinct spectra already rigidify the geometry ($[A, \Sigma_R] = 0$ forces $A = 0$). The rank-indexed rigidity law $\dim \operatorname{Aut} = 6, 2, 0, 0$ is a *degenerate-stratum* phenomenon (isotropic blocks) and is not exercised by generic data. This is consistent with, not contrary to, the framework: the same top-stratum / isotropic-stratum distinction governs the saturation of invariants (§6). The real system is maximally rigid, as a generic operating point should be.

6 The saturation correction manifests on real data

The recent correction—that canonical correlations under-determine a resolution geometry, and the frame-alignment invariants are needed—is not an abstract nicety; it shows up here.

Finding 6.1 (Frame alignment is needed on real data). Starting from the data coupling C , construct C' with the *same* canonical correlations (rotate the whitened coupling). Then $\rho(C') = \rho(C)$ to 10^{-6} , yet

$$\operatorname{tr}(\Sigma_R C \Sigma_N C^\top) = 3.63 \quad \text{while} \quad \operatorname{tr}(\Sigma_R C' \Sigma_N C'^\top) = 1.97.$$

The two geometries share every canonical correlation but differ in a gauge invariant; (λ, μ, ρ) would call them identical. On real data, as in theory, the frame-alignment invariants carry independent, necessary information.

7 Puzzle-closure summary

RG component	test on chaotic data	outcome
chaos of generator	largest Lyapunov exponent	$0.14 > 0$
resolved/null sectors	$\text{rank } J, R \perp N$	clean, $p = q = 6$
Schur floor	$\Sigma_N - C^\top \Sigma_R^{-1} C \succ 0$	holds, margin 0.25
canonical correlations	whitened SVD	graded 0.86–0.0003
frame-alignment invariants	m_{ab} present, $O(1)$	yes (3.63, 1.20)
gauge invariance	ρ, m_{ab} vs raw C	invariant / moves
index lock	$\text{rank } J$ along loop	stable at 6
atlas curvature (small loops)	holonomy vs area	$\propto$ area, SO
stratum walls (large loop)	singular-gap collapse	appears (4×10^{-3})
symmetry interlock	$\dim \text{Aut}$ at operating pt	0 (top-stratum rigid)
saturation correction	matched ρ , differing m_{ab}	confirmed

Every component computes; every consistency condition holds; and the two places where naive computation stalls (large-loop holonomy, the interlock) are precisely the stratified-geometry predictions. The puzzle closes.

8 Non-claims

Non-claim 8.1. No physical claim is made. The vortex system is a data generator; nothing about turbulence, Navier–Stokes, energy cascades, or scaling laws is asserted or implied. Those interpretations are retired.

Non-claim 8.2. The test establishes *computability and internal consistency* of the RG machinery on realistic data, not correctness of any external application. A different generator would exercise the same machinery; the point is that the machinery is well-posed and self-consistent, not that this system is special.

Non-claim 8.3. The atlas curvature is a property of the observation geometry (how the resolved subspace turns as controls vary), not a physical field. The holonomy is the geometric phase of the resolved subspace bundle.

9 Conclusion

Handed data it did not design—a chaotic Biot–Savart vortex field observed on a probe shell—resolution geometry computes end to end and stays consistent. Sectors, the Schur floor, a realistic graded spectrum of canonical correlations, the frame-alignment invariants, gauge invariance, and the index lock all hold. The atlas glues: on small top-stratum loops the resolved subspace transports with a genuine, area-scaling curvature; on the large loop it meets the stratum walls the geometry predicted, and the singular structure appears on its own. The symmetry interlock reads maximal rigidity, as a generic operating point should, and the saturation correction shows its necessity in the data. The pieces fit. What has been demonstrated is the internal coherence and computability of the framework on realistic input—no more, and no less.

A Numerical summary

`numpy` only; vortex integration by RK4 ($dt = 10^{-2}$, $T = 6$); Jacobian by finite differences ($\varepsilon = 10^{-4}$); covariance from an 80-member IC ensemble in the top-12 signal subspace with a

15% noise floor; loop transport by the discrete Grassmann Wilson loop.

- Lyapunov exponent ≈ 0.14 ; $\text{cond}(\Sigma) \approx 1.8 \times 10^2$; $p = q = 6$.
- Schur complement min eigenvalue 0.048 (margin/scale 0.25).
- $\rho = (0.859, 0.544, 0.459, 0.305, 0.040, 0.0003)$; $m_{00} = 1.20$, $m_{11} = 3.63$.
- Gauge: ρ, m_{ab} invariant to 10^{-7} ; $\|C_g - C\| = 1.73$.
- Small-loop holonomy angle/area = 3.26, 3.81, 3.23 for $r = 0.05, 0.10, 0.15$; $\det = 1$.
- Large loop: min singular gap 4×10^{-3} (wall); $\dim \text{Aut} = 0$ at the operating point.
- Saturation: matched ρ (10^{-6}), $m_{11} = 3.63$ vs. 1.97.

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